Grade 8 Name____________________
Unit 4 Date____________________
Lesson 6 Practice Period____________________

1. Write an exponential expression equivalent to 8^3 that has a base of 2.

2. Write an exponential expression equivalent to 25^3 that has a base of 5.

3. Write an exponential expression equivalent to 64^2 that has a base of 8.

Marquise says the expressions $\left(\frac{1}{8}\right)^3 \cdot 2^{-3}$ and $\left(\frac{1}{4}\right)^5 \cdot 2^{-2}$ are equivalent because both expressions can be rewritten as 2^{-12}.

In questions 4–5, generate an equivalent expression for each expression using a base of 2. Show all work.

4. $\left(\frac{1}{8}\right)^3 \cdot 2^{-3}$

5. $\left(\frac{1}{4}\right)^5 \cdot 2^{-2}$

6. Is Marquise correct? Explain your answer.

In each column, generate an equivalent expression using as few bases as possible.
Then match Column A expressions with expressions in Column B that are equivalent.

Column A	Work	Column B	Work
7. $2^3 \cdot 2^4$		A. $3^{-3} \cdot 2^6$	
8. $\frac{2^6}{3^3}$		B. $\frac{(3 \cdot 2)^{-2}}{3^{-4}}$	
9. 2^0		C. $\left(\frac{1}{2}\right)^{-3} \cdot \left(\frac{1}{2}\right)^{-4}$	
10. $\left(\frac{3}{2}\right)^2$		D. 1	

Question 7 in Column A is equivalent to Letter ______ in Column B.

Question 8 in Column A is equivalent to Letter ______ in Column B.

Question 9 in Column A is equivalent to Letter ______ in Column B.

Question 10 in Column A is equivalent to Letter ______ in Column B.